

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0059

Green Run 138 kV
June 25, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0059. The Interconnection Customer proposes a 97.9 MW generating facility interconnecting at the Green Run 138 kV substation in the Virginia area. The request was received on July 18, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 54 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$308,821,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Seco – Wing Lane 138.0 kV
Contingency: Haggin – Mitchell Dam 345.0 kV Line
Loading: 215.3% (Rating: 167 MVA)
DFAX: 8.5%
Pre-project loading: 210.3%
2. Facility: Seco – Harvard 138.0 kV
Contingency: Mitchell Dam – Haggin 345.0 kV
Loading: 184.5% (Rating: 248 MVA)
DFAX: 8.5%
Pre-project loading: 181.1%
3. Facility: Walthall Substation – Haggin Substation 138.0 kV
Contingency: N Tower Tap – Haggin 345.0 kV
Loading: 167.2% (Rating: 236 MVA)
DFAX: 6.4%
Pre-project loading: 164.5%
4. Facility: Carlin – Trabue 138.0 kV
Contingency: Mitchell Dam – Skaggs Island Junction 2 345.0 kV
Loading: 165.0% (Rating: 184 MVA)
DFAX: 5.8%
Pre-project loading: 161.8%
5. Facility: Fruitridge & S. Land Park – Carlin 138.0 kV
Contingency: Skaggs Island Jct 2 – Mitchell Dam 345.0 kV
Loading: 164.8% (Rating: 173 MVA)
DFAX: 5.8%
Pre-project loading: 161.5%

6. Facility: Trabue Substation – Duplainville 138.0 kV
Contingency: Skaggs Island Jct 2 – Mitchell Dam 345.0 kV
Loading: 156.5% (Rating: 206 MVA)
DFAX: 5.8%
Pre-project loading: 153.7%
7. Facility: Duplainville – Bank Of America 138.0 kV
Contingency: Skaggs Island Jct 2 – Mitchell Dam 345.0 kV
Loading: 154.9% (Rating: 226 MVA)
DFAX: 5.8%
Pre-project loading: 152.4%
8. Facility: Harvard – Hull St Road 138.0 kV
Contingency: Mitchell Dam – Haggin Substation 345.0 kV
Loading: 150.3% (Rating: 333 MVA)
DFAX: 8.5%
Pre-project loading: 147.8%
9. Facility: Wing ane – Spring St 138.0 kV
Contingency: Mitchell Dam – Haggin 345.0 kV
Loading: 129.4% (Rating: 208 MVA)
DFAX: 6.3%
Pre-project loading: 126.4%
10. Facility: Fritch – Lakeside 138.0 kV
Contingency: Mitchell Dam – Skaggs Island Jct 2 345.0 kV
Loading: 120.8% (Rating: 216 MVA)
DFAX: 6.1%
Pre-project loading: 118.0%
11. Facility: Bear Island – Dunn Siding 345.0 kV
Contingency: Leavitt – Birchwood 345.0 kV
Loading: 118.6% (Rating: 1,303 MVA)
DFAX: 24.3%
Pre-project loading: 116.8%
12. Facility: Fluvanna P.S – Louisa Generation Facility 345.0 kV
Contingency: Leavitt – Birchwood 345.0 kV
Loading: 114.3% (Rating: 1,711 MVA)
DFAX: 12.5%
Pre-project loading: 113.5%
13. Facility: Redbed Plains Wind Farm – Fruitridge & S. Land Park 138.0 kV
Contingency: Mitchell Dam – Skaggs Island Jct 2 345.0 kV
Loading: 113.1% (Rating: 219 MVA)
DFAX: 5.8%
Pre-project loading: 110.5%
14. Facility: Fritch – Big Hanaford 138.0 kV
Contingency: Mitchell Dam – Skaggs Island Junction 2 345.0 kV
Loading: 112.3% (Rating: 281 MVA)
DFAX: 8.6%
Pre-project loading: 109.3%
15. Facility: Unknown – N Tower Tap 138.0 kV
Contingency: N Tower Tap – Haggin Substation 345.0 kV
Loading: 109.4% (Rating: 277 MVA)
DFAX: 7.9%
Pre-project loading: 106.6%
16. Facility: Walthall – Unknown 138.0 kV
Contingency: North Tower Tap – Haggin 345.0 kV

- Loading: 106.2% (Rating: 272 MVA)
DFAX: 7.9%
Pre-project loading: 103.3%
17. Facility: Dist 1000 – Birchwood 138.0 kV Line
Contingency: BearIsland – Dunn Siding 345.0 kV line
Loading: 104.2% (Rating: 304 MVA)
DFAX: 5.6%
Pre-project loading: 102.4%
18. Facility: Carrabelle – Louisa Generation Facility 345.0 kV
Contingency: Louisa Generation Facility – North Anna 345.0 kV
Loading: 103.9% (Rating: 1,492 MVA)
DFAX: 6.0%
Pre-project loading: 103.5%
19. Facility: Louisa Generation Facility – N Anna 345.0 kV
Contingency: Louisa Generation Facility – Carrabelle 345.0 kV
Loading: 102.7% (Rating: 1,686 MVA)
DFAX: 5.9%
Pre-project loading: 102.3%
20. Facility: Lakeside – Natchez 138.0 kV line
Contingency: Mitchell Dam – Skaggs Island Jct 2 345.0 kV
Loading: 102.4% (Rating: 197 MVA)
DFAX: 6.1%
Pre-project loading: 99.4%
21. Facility: North Tower Tap 345-138 kV Auto
Contingency: N Tower Tap – Haggin 345.0 kV
Loading: 101.5% (Rating: 438 MVA)
DFAX: 12.0%
Pre-project loading: 98.8%
22. Facility: Bear – Cloer 345.0 kV line
Contingency: Bear – Mecklenburg Power Sta 345.0 kV
Loading: 101.2% (Rating: 2,416 MVA)
DFAX: 7.0%
Pre-project loading: 100.9%
23. Facility: Trenton Diesel – Dist 000 138.0 kV Line
Contingency: Dunn Siding – Bear Island Substation 345.0 kV
Loading: 100.7% (Rating: 325 MVA)
DFAX: 5.6%
Pre-project loading: 99.0%
24. Facility: Spring St – Chipola 138.0 kV
Contingency: Mitchell Dam – Haggin 345.0 kV
Loading: 100.3% (Rating: 234 MVA)
DFAX: 6.3%
Pre-project loading: 97.6%
25. Facility: Jim Hill – Haggin 138.0 kV L
Contingency: Skaggs Island Jct 2 – Mitchell Dam 345.0 kV Line
Loading: 100.3% (Rating: 257 MVA)
DFAX: 5.6%
Pre-project loading: 98.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Seco – Wing Lane 138.0 kV	Rebuild line with higher-capacity conductor	\$2,097,000	100% (\$2,097,000)
Seco – Harvard 138.0 kV	Rebuild line with higher-capacity conductor	\$9,067,000	100% (\$9,067,000)
Walthall – Haggin 138.0 kV	Rebuild line with higher-capacity conductor	\$5,649,000	100% (\$5,649,000)
Carlin – Trabue 138.0 kV	Rebuild line with higher-capacity conductor	\$2,182,000	100% (\$2,182,000)
Carlin Substation – Fruitridge & S. Land Park 138.0 kV	Rebuild line with higher-capacity conductor	\$1,262,000	100% (\$1,262,000)
Trabue – Duplainville 138.0 kV	Rebuild line with higher-capacity conductor	\$3,459,000	100% (\$3,459,000)
Bank Of America – Duplainville 138.0 kV	Rebuild line with higher-capacity conductor	\$1,166,000	100% (\$1,166,000)
Harvard – Hull St Road 138.0 kV	Rebuild line with higher-capacity conductor	\$1,388,000	100% (\$1,388,000)
Wing Lane – Spring Street 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,847,000	100% (\$2,847,000)
Fritch – Lakeside 138.0 kV	Rebuild line with higher-capacity conductor	\$3,246,000	100% (\$3,246,000)
Bear Island – Dunn Siding 345.0 kV	Rebuild line with higher-capacity conductor	\$46,697,000	100% (\$46,697,000)
Fluvanna P.S – Louisa Generation Facility 345.0 kV	Rebuild line with higher-capacity conductor	\$72,614,000	100% (\$72,614,000)
Redbed Plains Wind Farm – Fruitridge & S. Land Park 138.0 kV	Rebuild line with higher-capacity conductor	\$11,973,000	100% (\$11,973,000)
Fritch – Big Hanaford 138.0 kV	Rebuild line with higher-capacity conductor	\$4,906,000	100% (\$4,906,000)
Unknown – N Tower Tap 138.0 kV	Reconductor line	\$3,241,000	100% (\$3,241,000)
Walthall – Unknown 138.0 kV	Reconductor line	\$6,729,000	100% (\$6,729,000)
Dist 1000 – Birchwood 138.0 kV Line	Reconductor line	\$6,883,000	100% (\$6,883,000)
Carrabelle – Louisa Generation Facility 345.0 kV	Reconductor line	\$42,147,000	100% (\$42,147,000)
North Anna – Louisa Generation Facility 345.0 kV	Reconductor line	\$20,388,000	100% (\$20,388,000)
Lakeside – Natchez 138.0 kV	Reconductor line	\$1,067,000	100% (\$1,067,000)
North Tower Tap 345-138 kV Auto	Replace transformer with higher-rated unit	\$16,885,000	100% (\$16,885,000)
Bear – Cloer 345.0 kV line	Reconductor line	\$27,429,000	100% (\$27,429,000)
Dist 1000 – Trenton Diesl 138.0 kV L	Reconductor line	\$12,046,000	100% (\$12,046,000)
Spring St – Chipola 138.0 kV	Reconductor line	\$3,129,000	100% (\$3,129,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jim Hill – Haggin 138.0 kV L	Reconductor line	\$324,000	100% (\$324,000)
	Total allocated to project		\$308,821,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.